

SUPRIYA BHOWMICK

Software Engineer | Full-Stack Developer | AI / GenAI Specialist

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PROFESSIONAL SUMMARY

B.Tech graduate in Information Technology (Narula Institute of Technology, 2025 — GPA: 8.13) who builds things that work in production, not just in theory. Strong in Data Structures & Algorithms (400+ problems across LeetCode, GeeksforGeeks, HackerRank) with the engineering instincts to translate that rigour directly into scalable system design and clean architecture decisions. Independently architected and delivered a parallel multi-agent GenAI platform — LangGraph state machine orchestrating Gemini, Mistral, and Cohere with concurrent execution, Zod-validated contracts, and a pipeline designed to scale without restructuring. Fluent across the full modern stack: React.js, TypeScript, Node.js, REST APIs, MongoDB, AWS (EC2, Lambda, S3, RDS). Specialises in GenAI engineering — LangChain, LangGraph, multi-agent orchestration, prompt engineering, and AI-native development workflows using Claude Code, GitHub Copilot, and Cursor. Brings the ownership mindset and technical depth of a senior engineer, with the hunger and adaptability of someone just getting started.

CORE COMPETENCIES

- Full-Stack Web Development
- React.js · TypeScript · Node.js
- REST API Design & Architecture
- MongoDB · MySQL · Redis
- CI/CD · Git Workflows · Agile SDLC
- AI / GenAI Integration & LLM Orchestration
- LangChain · LangGraph · Multi-Agent Systems
- AWS Cloud Infrastructure (EC2, Lambda, S3, RDS)
- System Design · Scalable Architecture
- Prompt Engineering · Structured Output Validation (Zod)

TECHNICAL SKILLS

Frontend	React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, SCSS, Redux, Context API, MERN Stack
AI / GenAI	LangChain, LangGraph, OpenAI APIs, Multi-Agent Systems, LLM Orchestration, Prompt Engineering, AI Workflow Design
AI Dev Tools	Claude Code, GitHub Copilot, Cursor, ChatGPT — code generation, refactoring, debugging, documentation
Backend	Node.js, Express.js, REST APIs, Next.js
Cloud & DevOps	AWS (EC2, Lambda, RDS, S3), Git, GitHub, Postman, CI/CD basics
Databases	MySQL, MongoDB, Redis
Testing	Postman (API testing), Jest (basic), Manual Frontend QA
CS Fundamentals	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Languages	C, C++, JavaScript, TypeScript, Python

PROFESSIONAL EXPERIENCE

Software Engineering Consultant (Virtual) | Deloitte Australia

Apr 2025 – Jun 2025

IIoT Data Engineering & Enterprise Dashboard Development

- Engineered a Python data transformation pipeline processing heterogeneous IIoT machine telemetry into standardized JSON, enabling reliable downstream analytics across enterprise systems.
- Architected an enterprise dashboard proposal in React, Tailwind CSS, and Redux — covering system architecture, multi-sprint timeline, cost-benefit analysis, and horizontal scalability strategy.
- Drove end-to-end SDLC adherence: requirements, design, implementation, code review, and technical documentation mirroring production engineering workflows.
- Integrated AI-assisted development (GitHub Copilot, Cursor) to accelerate delivery, validate logic, and enforce code quality across the project lifecycle.

KEY PROJECTS

AI Multi-Model Judge System | Production GenAI Platform

[GitHub](#)

TypeScript · Node.js · Express.js · LangChain · LangGraph · Gemini · Mistral · Cohere · Zod · React

- Sole architect and engineer of a multi-agent AI evaluation platform: Mistral and Cohere compete in parallel; Gemini acts as an intelligent judge rendering scored, reasoned verdicts.
- Designed a deterministic LangGraph state machine (User Input → Express API → runGraph() → solutionNode [parallel] → judgeNode → structured JSON) — eliminating ad-hoc prompt chaining for reproducible, testable AI workflows.
- Reduced response latency through concurrent LLM execution via LangGraph's async model, replacing serial API calls with parallel invocations.
- Enforced type-safe AI outputs across three LLM providers using Zod schema validation — guaranteeing zero-failure JSON contracts between backend and frontend.
- Architected for open/closed extensibility: new AI providers plug in without pipeline restructuring, enabling rapid ensemble expansion and provider A/B testing.

Notes REST API | Full-Stack Cloud-Deployed Application

[GitHub](#)

Node.js · Express.js · MongoDB · React · Axios · Postman · Render

- Delivered a production-deployed full-stack Notes app on Render: RESTful API (GET/POST/PATCH/DELETE), CRUD-complete MongoDB integration, React/Axios frontend — fully tested via Postman.

Sorting Algorithm Visualizer | Interactive Algorithm Education Tool

[GitHub](#)

JavaScript · HTML · CSS · React

- Built a real-time step-by-step visualization engine for Bubble Sort, Merge Sort, and Quick Sort — translating computational complexity into an interactive, fully responsive learning interface.

EDUCATION

B.Tech, Information Technology | Narula Institute of Technology, Kolkata

2021 – 2025 | GPA: 8.13 / 10

CERTIFICATIONS & ACHIEVEMENTS

- Data Structures & Algorithms with C / C++ | Certified | [View Certificate](#)
- Full Stack Web Development | Certified | [View Certificate](#)
- Solved 400+ coding problems across LeetCode, GeeksforGeeks, and HackerRank — arrays, trees, graphs, dynamic programming, and number theory.
- Ranked active competitor on HackerRank (110+ problems) and HackerEarth with consistent contest participation and daily challenge streaks.